Datamites

Global Institute for Data Science

CUSTOMER SATISFACTION (CSAT) ANALYSIS

POWER BI DASHBOARD
CAPSTONE PROJECT REPORT

Submitted as Part of Internship at Datamites

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ABSTRACT

Customer Satisfaction Analytics (CSAT) is one of the most important performance indicators used by organizations to understand customer experience and service quality. In today's competitive environment, organizations continuously interact with customers through different communication channels such as phone calls, emails, chat support, and social media platforms. Measuring customer satisfaction helps organizations understand whether customer expectations are being fulfilled.

The objective of this project is to analyze customer interaction data and derive meaningful insights from customer satisfaction scores, service channels, customer response behavior, and issue categories. The project uses data analytics techniques to clean, process, transform, and visualize customer-related information.

Interactive dashboards were created using visualization tools to monitor customer trends, identify service gaps, and support better decision-making. The analysis helps management understand customer needs, improve service quality, and increase customer retention.

INTRODUCTION

Customer satisfaction directly affects business growth, customer loyalty, and brand reputation. Companies that understand customer expectations and continuously improve their services usually perform better than competitors.

Organizations collect customer data from multiple sources such as:

* Customer support calls
* Feedback surveys
* Emails and chat systems
* Complaint management systems
* Social media interactions

However, raw customer data alone does not provide meaningful insights. Analytics techniques are required to transform large datasets into understandable information.

Customer Satisfaction Analytics helps businesses:

* Measure customer happiness levels
* Monitor service quality performance
* Identify customer pain points
* Understand customer interaction patterns
* Improve operational efficiency

This project focuses on analyzing customer service data using business intelligence tools and generating actionable insights through dashboards and visualizations.

PROBLEM STATEMENT

Organizations receive large volumes of customer interactions daily. Manually analyzing this information becomes difficult due to:

* Large data size
* Multiple communication channels
* Different customer behaviors
* High number of service requests
* Inconsistent customer feedback

Without proper analytics:

* Service issues remain unidentified
* Customer dissatisfaction increases
* Response quality decreases
* Customer retention becomes difficult

Therefore, an analytics solution is required to monitor customer satisfaction metrics and support data-driven decisions

PROJECT OBJECTIVES

The major objectives of this project are:

Primary Objectives:

* Analyze customer satisfaction levels
* Measure service performance
* Study customer interaction behavior
* Monitor call and service trends

Secondary Objectives:

* Create interactive dashboards
* Identify low-performing areas
* Compare customer channels
* Generate recommendations
* Improve customer experience

Business Objectives:

* Increase customer retention
* Improve service quality
* Reduce customer complaints
* Support management decision-making

DATASET DESCRIPTION

The dataset contains customer interaction information collected from customer support operations.

Dataset Attributes:

Customer ID:

Unique identification number assigned to customers.

Call Date:

Date on which customer interaction occurred.

Call Channel:

Communication method used by customers such as:

* Phone

* Email

* Chat

* Web

CSAT Score:

Customer satisfaction rating given after service interaction.

Response Time:

Time taken to respond to customer queries.

Call Duration:

Total interaction duration.

Issue Category:

Category of customer problem or request.

State / Region:

Customer location information.

Resolution Status:

Whether customer issue was resolved or unresolved.

Feedback Comments:

Customer-provided suggestions or complaints.

Dataset Importance:

The dataset helps understand:

* Customer behavior
* Service performance
* Regional patterns
* Communication preferences
* Satisfaction drivers

TOOLS AND TECHNOLOGIES USED

Power BI:

Used for:

* Dashboard creation
* Interactive visualizations
* KPI cards
* DAX calculations

Excel:

Used for:

* Data understanding
* Initial inspection
* Data validation

Power Query:

Used for:

* Data transformation
* Removing duplicates
* Cleaning records

SQL:

Used for:

* Data extraction
* Filtering datasets
* Query operations

DAX:

Used for creating:

* Measures

* Calculated columns

* KPIs

METHODOLOGY / PROJECT PROCESS

Phase 1: Data Collection

Customer interaction data was collected from structured datasets containing service information.

Phase 2: Data Understanding

Performed:

* Column analysis
* Data type checking
* Missing value inspection

Phase 3: Data Cleaning

Cleaning activities included:

* Removing duplicates
* Handling null values
* Standardizing formats
* Correcting errors

Phase 4: Data Transformation

Created:

* New calculated columns
* Date hierarchy
* Measures and KPIs

Phase 5: Data Modeling

Established relationships among tables for efficient analysis.

Phase 6: Dashboard Development

Created:

* KPI cards
* Trend charts
* Maps
* Donut charts
* Bar charts

Phase 7: Analysis and Insights

Performed business analysis to identify trends and recommendations.

DATA CLEANING AND PREPROCESSING

Data preprocessing is essential because poor-quality data affects analytics accuracy.

Cleaning Steps:

1. Removed duplicate records
2. Handled missing values
3. Corrected date formats
4. Standardized text values
5. Removed unnecessary columns
6. Corrected invalid entries
7. Converted data types
8. Created calculated columns

Benefits:

* Increased data accuracy
* Reduced inconsistencies
* Improved dashboard performance

DASHBOARD DEVELOPMENT

KPIs Created:

* Total Calls

* Average CSAT Score

* Customer Satisfaction %

* Average Response Time

* Total Resolved Cases

Charts Used:

Line Chart:

Used to visualize call trends over time.

Donut Chart:

Shows distribution of customer communication channels.

Map Visualization:

Displays state-wise customer distribution.

Bar Charts:

Compare categories and performance metrics.

Slicers:

Provide dynamic filtering options.

Dashboard Benefits:

* Better monitoring

* Faster decisions

* Interactive reporting

KEY INSIGHTS

Major insights derived:

* Certain regions generated more customer interactions.

* Phone support handled most customer requests.

* Higher response times reduced satisfaction scores.

* Some issue categories produced repeated complaints.

* Peak periods showed increased customer volume.

* Faster issue resolution improved customer ratings.

RECOMMENDATIONS

Based on analysis:

1. Improve slow response areas.
2. Increase staffing during peak hours.
3. Improve low-performing service channels.
4. Provide customer support training.
5. Monitor recurring complaints.
6. Introduce automated customer support systems.
7. Track customer satisfaction continuously.

Expected Benefits:

* Improved customer retention
* Better service quality
* Reduced complaints
* Increased operational efficiency

CONCLUSION

The Customer Satisfaction Analytics project successfully transformed raw customer data into meaningful business insights.

Through data cleaning, visualization, and analysis, the project identified service gaps, customer preferences, and performance trends.

Interactive dashboards help organizations monitor customer experience and make data-driven decisions.

This project demonstrates the importance of analytics in improving customer satisfaction and business performance.

FUTURE SCOPE

Future enhancements may include:

* Real-time dashboard integration
* Predictive customer analytics
* Sentiment analysis models
* Machine learning integration
* Automated alert systems
* AI-based customer support analytics

These improvements can further increase analytical capability and business value.

REFERENCES

1. Microsoft Power BI Documentation. Retrieved from: https://learn.microsoft.com/power-bi/

2. Microsoft Power Query Documentation. Retrieved from: https://learn.microsoft.com/power-query/

3. Microsoft DAX Documentation. Retrieved from: https://learn.microsoft.com/dax/

4. Datamites Internship Materials and Project Guidelines.

5. Dataset provided for Customer Satisfaction Analytics project.

6. Microsoft Learn Resources for Power BI Dashboard Development.

7. Business Intelligence concepts and visualization techniques used during project implementation.